Условной плотностью $\xi|\eta$ наз-ся $p_{\xi|\eta}(x|y)\geq 0$:

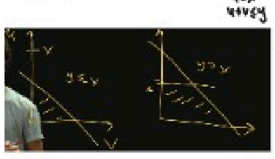
$$\boxed{P(\xi\leq x|\eta=y)=\int_{-\infty}^{x}p_{\xi|\eta}(t,y)\,dt}$$

Утв| Если (ξ,η) - абс. непр., т.е. $\exists\, p_{(\xi,\eta)}(x,y)$ =>

$$\Rightarrow \exists\, p_{\xi|\eta}(x,y)=\begin{cases}\dfrac{p_{(\xi,\eta)}(x,y)}{p_\eta(y)}, & p_\eta(y)\neq 0\\ 0, & p_\eta(y)=0\end{cases}$$

$p_{(X,X+Y)}(x,y)=?$

$$F_{(X,X+Y)}(x,y)=P(X\leq x, X+Y\leq y)=\int\limits_{\substack{u\leq x\\ u+v\leq y}} e^{-u}I(u\geq 0)e^{-v}I(v\geq 0)\,du\,dv=$$



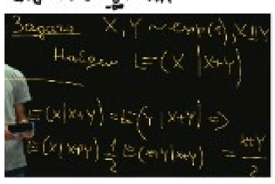
$$y\leq x \Rightarrow \int_0^y\int_0^{y-u}e^{-u}e^{-v}\,du\,dv=\int_0^y e^{-u}(1-e^{u-y})\,du=$$

$$=\int_0^y e^{-u}du-\int_0^y e^{-y}du=2e^{-y}$$

$$y>x \Rightarrow \int_0^x e^{-u}du\int_0^{y-x}e^{-v}dv=\int_0^x e^{-u}(1-e^{y-x})\,du=\int_0^x(e^{-u}-e^{y-u-x})\,du$$

Ф-ла УМО в непрерывном случае:

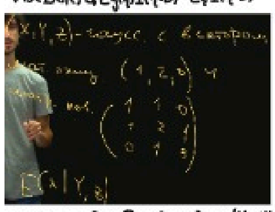
$$E(g(\xi)|\eta=y)=\int_{-\infty}^{+\infty}g(x)\,p_{\xi|\eta}(x|y)\,dx$$



ξ - сл. вектор, η - сл. в-на

$E(\xi|\eta)=g(\eta)$

$\forall B\in\mathcal{B}(\mathbb{R}) \;\llcorner\; Eg(\eta)I(\eta\in B)=E\xi I(\eta\in B)$



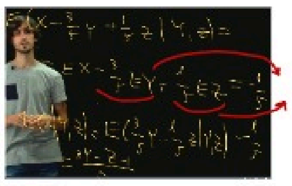
перейдем к поиску независимых в-ров $(X-\alpha Y-\beta Z, Y)$, $(X-\alpha Y-\beta Z, Z)$

$$cov(X-\alpha Y-\beta Z, Y)=cov(X,Y)-\alpha\,cov(Y,Y)-\beta\,cov(Z,Y)=0$$
$$1-2\alpha-\beta=0$$
$$cov(X-\alpha Y-\beta Z, Z)=cov(X,Z)-\alpha\,cov(Y,Z)-\beta\,cov(Z,Z)=0$$
$$0-\alpha-3\beta=0$$

$$\begin{cases}1-2\alpha-\beta=0\\ -\alpha-3\beta=0\end{cases}\qquad 1+6\beta-\beta=0\qquad 5\beta=-1$$

$\alpha=-3\beta$ => $\underline{\beta=-\frac{1}{5}}$ => $\underline{\alpha=\frac{3}{5}}$

$$\Rightarrow E(\underbrace{X-\alpha Y-\beta Z}_{}|\underbrace{Y,Z}_{\perp\!\!\!\perp})=E(X-\alpha Y-\beta Z)=EX-\alpha EY-\beta EZ=1-2\alpha=1-\tfrac{6}{5}=-\tfrac{1}{5}$$



$$E(X|Y,Z)=\underbrace{E(X-\tfrac{3}{5}Y+\tfrac{1}{5}Z|Y,Z)}_{=-\frac{1}{5}}+\underbrace{E(\tfrac{3}{5}Y|Y,Z)}_{=\frac{3}{5}\cdot 2=\frac{6}{5}}-\underbrace{E(\tfrac{1}{5}Z|Y,Z)}_{=0}=1$$

$$E(\xi|\mathcal{G}),\ \mathcal{G}\subset\mathcal{F}$$

сл. в. на $(\Omega,\mathcal{G},P)$:

$\forall A\in\mathcal{G} \;\llcorner\; E(E(\xi|\eta)I_A)=E\xi I_A$

η - сл. в.

$\sigma(\eta)=\{\eta^{-1}(B), B\in\mathcal{B}(\mathbb{R})\}$

$E(\xi|\eta)=E(\xi|\sigma(\eta))$

в условии м. поставить сл. в-ры, сигма-алгебру и др.